

Parisara-Cycle

Green Commuter Guide - User Guide

Purpose: Parisara-Cycle helps bicycle commuters choose safer routes, report danger zones, find basic pit-stops, track CO2 savings, and optionally share location with nearby riders.

Current mode: The app works without Firebase or Google API keys. It uses local login, local map display, local route generation, mock pit-stops, and demo buddy markers. Real Firebase and Google routing can be connected later.

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1. Starting the App

1. Install and open the Parisara-Cycle app on the emulator or Android device.
2. The first screen is the login screen.
3. If location permission is requested, you can allow it for current-location features. The app still works in demo mode if you deny it.

2. Login or Register

Use: Creates a rider session so app data can be linked to a user.

1. Enter any valid email address.
2. Enter a password with at least 6 characters.
3. Tap **Register** for first-time use, or **Sign in** after that.

In the current no-key mode, login is stored locally on the device. If Firebase is added later, the same screen uses Firebase Email/Password Authentication.

3. Bottom Navigation

The app has three main tabs at the bottom:

- **Map:** Routes, danger zones, pit-stops, and buddy markers.
- **Stats:** Daily and monthly CO2 savings.
- **Profile:** Account details, live-location sharing, and sign out.

4. Safe Route Map

Use: Helps cyclists select a source and destination, then generate a bicycle-friendly route.

1. Open the **Map** tab.
2. Tap **Allow location** if you want the app to use your current location as the route source.
3. If location is not allowed, the app uses a default demo source point.
4. Tap **Pick destination**.
5. Tap anywhere on the map to set the destination.
6. Tap **Route** to draw the route.

The route distance is used to calculate CO2 savings. In no-key mode, the app generates a local bicycle-style route. With Google Directions configured, it can use real bicycle routing.

5. Danger Zone Reporting

Use: Lets riders mark unsafe places so they can be avoided or watched carefully.

1. Open the **Map** tab.
2. Long-press on the map where the danger exists.
3. Select one marker type:
 - **Pothole:** Road damage that may cause a fall or tire damage.
 - **Dangerous Intersection:** Junctions with high traffic risk or poor visibility.
 - **Blocked Path:** Construction, parked vehicles, debris, or any blocked cycling path.
4. The marker appears on the map.

In no-key mode, danger zones are saved locally. With Firebase configured, they are saved to Firestore with latitude, longitude, type, timestamp, and user ID.

6. Pit-stop Finder

Use: Shows helpful stops near the cycling area.

- **Cycle repair shops:** Useful for punctures, brake issues, chain repair, and basic service.
- **Water points:** Useful for hydration during the ride.

Open the **Map** tab and keep the **Pit-stops** filter enabled. Pit-stop markers are shown automatically near the current map area.

The current version uses mock pit-stop data so the feature works without Places API billing or setup.

7. Eco-stats

Use: Tracks the environmental benefit of cycling.

The app uses this rule:

1 km cycled = 120 g CO2 saved

1. Create a route from the **Map** tab.
2. After the route is generated, the distance is added to your stats.
3. Open the **Stats** tab.
4. View:
 - **Today:** Distance cycled and CO2 saved today.
 - **This month:** Monthly cycling distance and CO2 saved.

Stats are stored locally on the device.

8. Buddy System

Use: Helps cyclists see nearby riders and share live location.

1. Open the **Profile** tab.
2. Turn on **Live location**.
3. Return to the **Map** tab.
4. Keep the **Buddies** filter enabled to see nearby rider markers.

In no-key mode, the app shows demo nearby riders. With Firebase Realtime Database configured, it syncs live rider locations.

For battery efficiency, live location updates are designed to run at a controlled interval rather than constantly updating every second.

9. Profile

Use: Manages rider account and live sharing.

- View signed-in email and user ID.
- Enable or disable live-location sharing.
- Request location permission if needed.
- Tap **Sign out** to return to the login screen.

10. Map Filters

The Map screen includes filter chips:

- **Danger:** Shows or hides danger-zone markers.
- **Pit-stops:** Shows or hides repair and water markers.
- **Buddies:** Shows or hides nearby rider markers.
- **Route:** Shows or hides the selected route line.

11. Permissions

- **Location:** Used to set your current source point and share live location.

- **Internet:** Used later for Firebase, Google Directions, and online services. The no-key demo can still run without real backend setup.

12. Troubleshooting

App opens slowly first time: The emulator may take a few seconds to load Compose and Android services on first launch.

Map is not Google Maps: This is expected in no-key mode. The app uses a keyless local map so it can run without manual API setup.

No current location: Allow location permission and make sure the emulator/device location service is enabled.

Firebase data is not syncing: Add `app/google-services.json` and enable Firebase Authentication, Firestore, and Realtime Database.

Real bicycle routing is not available: Add valid Google Maps/Directions API keys to `local.properties`.

13. Quick Demo Flow

1. Open the app.
2. Register with an email and password.
3. Go to **Map**.
4. Tap **Pick destination**, then tap the map.
5. Tap **Route**.
6. Long-press the map and report a **Pothole**.
7. Open **Stats** to see CO2 savings.
8. Open **Profile**, turn on **Live location**, then return to **Map** to see buddy markers.